

Experiment No. 4

Aim: ML Modeling & Experiment Tracking

Objective: Build ML pipeline, tune hyperparameters, track experiments with MLflow.

Detailed Steps

1. Dataset Preparation

- Split data into train/test

2. Baseline Model Training

- Train a few baseline models (4-5 models)
- Evaluate using accuracy (classification) or RMSE (regression).

3. Hyperparameter Tuning

- Apply **GridSearchCV** or **RandomizedSearchCV** on one/two selected models.
- Compare tuned vs. baseline performance.

4. Experiment Tracking with MLflow

- Initialize MLflow run (`mlflow.start_run()`).
- Log hyperparameters, metrics, and model artifacts (`mlflow.log_param()`, `mlflow.log_metric()`, `mlflow.sklearn.log_model()`).
- View experiment results on MLflow UI dashboard.

5. Model Selection & Saving

- Select the best-performing model/models.
- Save as a serialized artifact (`.pkl` or MLflow model format).

Open-Source Tools

Scikit-learn, MLflow, Jupyter Notebook

Deliverables

- Trained ML models.
- Comparative analysis of baseline vs. tuned models.
- MLflow dashboard logs (runs, metrics, artifacts).

Conclusion

